## Affine-query recovery comparison on one RTX 4090

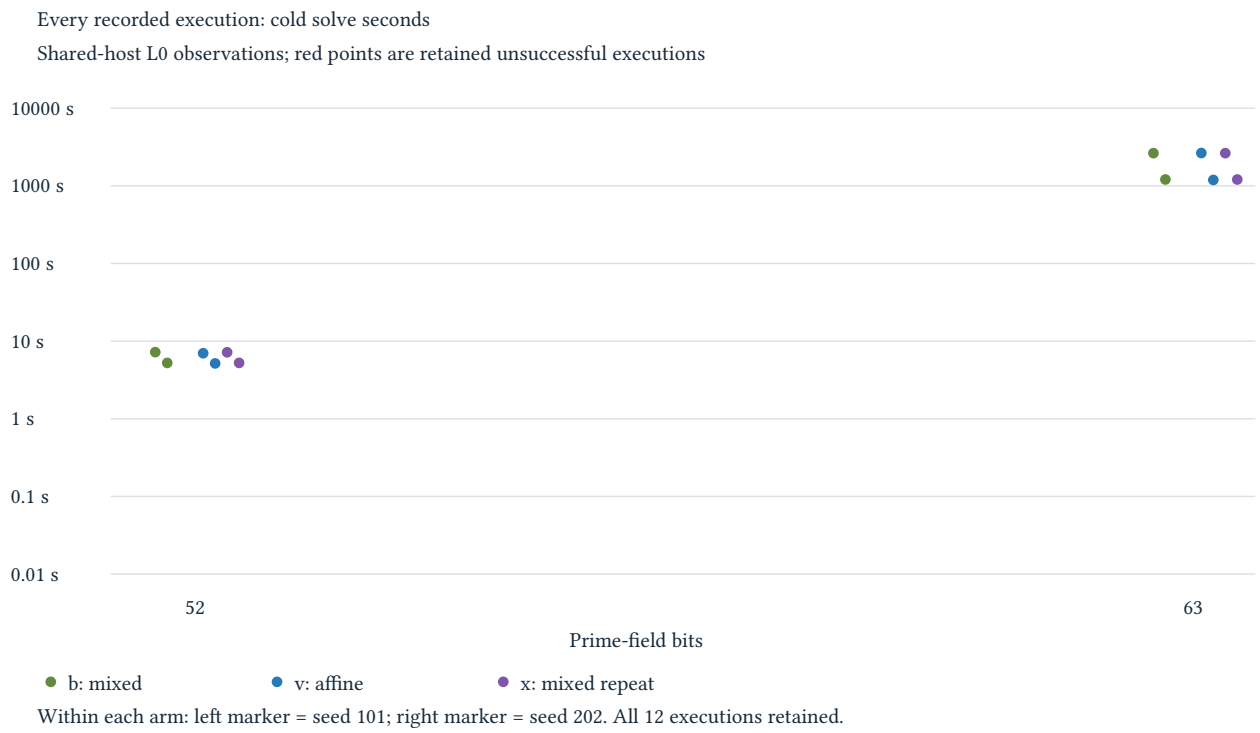
The opt-in affine-input subtraction was compared with the existing mixed formula on identical public targets, explicit bases, cache layouts and limits. Recorded executions: 12; verified recoveries: 12. Each successful result passed fresh CPU certificate replay and scalar+1 rejection. Complete certificates match byte for byte across formulas and repeats. Affine cold time was lower for both 52-bit targets and the 63-bit seed 202 target, and slightly higher for the 63-bit seed 101 target. The option remains off by default because the large-target effect was not uniform.

| Bits | Seed | b: mixed | v: affine | x: mixed repeat |
|---|---|---|---|---|
| 52 | 101 | 7.213224 / succeeded | 6.979429 / succeeded | 7.177102 / succeeded |
| 52 | 202 | 5.238853 / succeeded | 5.163135 / succeeded | 5.241295 / succeeded |
| 63 | 101 | 2636.991553 / succeeded | 2644.849780 / succeeded | 2636.255738 / succeeded |
| 63 | 202 | 1207.318937 / succeeded | 1190.595553 / succeeded | 1206.938904 / succeeded |

Cold solve seconds and native status, rounded from rows.csv. All rows are retained. The solve interval includes target generation, process startup, validation, base/cache construction, relation collection, elimination, extraction, internal replay, serialization and teardown. Fresh replay and negative-control time are included separately in wall_seconds.

| Bits | Seed | Candidate / mean(b,x) | AA spread / mean(b,x) |
|---|---|---|---|
| 52 | 101 | 0.970017 | 0.005020 |
| 52 | 202 | 0.985317 | 0.000466 |
| 63 | 101 | 1.003120 | 0.000279 |
| 63 | 202 | 0.986304 | 0.000315 |

Ratios are the frozen analyzer's outputs: candidate_seconds / mean(b,x) and abs(b-x) / mean(b,x), separately per fixed target. Values below 1 in the candidate column mean lower observed cold time. These L0 shared-host times are descriptive; the repeat spread is not a confidence interval. No population or cross-panel timing inference is made. Ratios require all three verified results.




Every execution from rows.csv is plotted separately, including any explicit limit or timeout. Two fixed targets at each prescribed field size.

## Formula scope, checks and custody

The source uses the established EFD mmadd-2007-bl affine-input formula: 4M+2S instead of 7M+4S in the nonexceptional addition formula. These are source formula operation counts, excluding conversions, normalization, lookup and memory traffic. Ordinary affine cache storage is unchanged. Infinity, inverses, doubling and zero-y doubling have explicit branches. Partial-cache and direct sum fallbacks retain mixed arithmetic; all three search entry points share one residual helper and preserve exact minimum witnesses.

Before data: 29 Rust tests, host/device differentials, Clippy, the original 27 query sanitizer checks and 12 affine-specific checks passed. The affine arithmetic fixture checked 37,584 cases on both CPU and GPU against an independent ordinary coordinate affine law, including exhaustive small curves and 63-bit cases. The affine packed/unpacked/persistent query paths passed memcheck, racecheck and synccheck; full service, certificate and cap-header checks also passed.

Command: bash research/explicit_relations_affine_20261030/run.sh. Fields: 52 and 63 bits; seeds: 101 and 202; 12 slots; R=ceil(r^(1/12)), tau=ceil(7R/5)+1. Query budget: 128 times (columns+1); maximum 16 trials. Caps: rows 2^32, work 2^34, chunk 2^20, columns 16384; reserve 25%. Order: b/v/x for seed 101, x/v/b for seed 202 at each size. Baseline repeat uses the exact baseline binary. Every successful cell must have a complete right cache and nonzero filter. No sufficient paper threshold guarantee is assumed for these explicit bases.

One NVIDIA GeForce RTX 4090, 24,564 MiB, driver 595.91.07, CC 8.9; CUDA 13.2/sm_89, Rust 1.88.0. Pod hnvmwjamem9ox9; provider rate USD 0.89/hour. Per-solve caps: 180/7200 s; replay: 60 s; fixture: 180 s; worker: 18000 s; controller: 18300 s. Periodic artifact copies every 60 s preserve evidence, not process restart. CPU user/system time and peak VRAM were not measured. Admission bytes are bounds, not observed peak usage. Build/preflight and shared curve generation are outside solve intervals. The code uses generic decomposition; no external-tool timing comparison was prescribed for this formula-only diagnostic.

Source: 333712634ae2fa0694236a530b7a5efd1be3efc5. Protocol SHA256: 2cfc1be77efbdc0d65031c1a42a19a72aca407aa69459e4440731993a92dad66 Archive SHA256: 4758f67ba19c0c12fe119f28404fd3c94b6c3dae481c630700d7ac0b179b24ba Raw records pushed as c79bd38363f9de485087a0e206032cf5938dbdae before this report. archive-manifest.json hashes every raw certificate/log in the approved S3 archive under validation/ecbench-explicit-x/affine-20261030-v1/results.tgz. Controller, bootstrap and validation exited 0; final archive integrity and pod deletion with fresh provider-inventory absence were verified. The native analyzer checks coverage, outcomes and equal relation/query counts; its malformed-input tests are documented in README.md. Canonical validation results are preserved with the raw export. Neither software checks nor this report promote an independent scientific-review verdict.

Primary formula source: https://www.hyperelliptic.org/EFD/g1p/auto-shortw-jacobian.html. Implementation assumptions and fixtures: gpu/explicit-relations/AFFINE-QUERY.md.